

HONDO, TX, USA

WMO: 722533

Lat: 29.360N

Lon: 99.174W

Elev: 920

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.6	30.7	10.7	9.8	40.0	14.8	12.1	43.8	27.2	55.3	24.5	57.8	6.4	0	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.2	100.5	73.3	99.0	73.5	97.1	73.4	78.2	89.4	77.5	88.4	76.9	87.5	10.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	138.1	81.0	75.0	135.8	80.7	74.2	132.4	80.3	42.3	89.9	41.5	88.5	40.8	87.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.8	19.6	17.7	20.1	103.5	3.7	3.2	17.5	105.8	15.3	107.7	13.2	109.6	10.5	111.9		
			18.0	79.8	3.7	0.9	15.3	80.4	13.2	80.9	11.1	81.4	8.5	82.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.7	52.0	56.6	63.0	69.9	77.0	82.6	84.2	85.0	79.9	71.2	60.9	52.9
	DBStd	13.70	8.64	9.66	8.38	7.12	5.77	3.60	3.30	3.17	5.01	7.76	8.78	8.57
	HDD50	213	79	41	9	0	0	0	0	0	0	1	16	67
	HDD65	1452	410	265	139	34	3	0	0	0	1	34	178	388
	CDD50	7389	141	225	412	598	837	977	1060	1085	897	659	342	156
	CDD65	3152	7	28	77	182	376	527	595	620	447	228	54	11
	CDH74	35036	81	255	622	1623	3644	6081	7312	7943	4864	2121	402	88
	CDH80	17318	9	59	165	577	1595	3127	4023	4533	2388	758	75	9
Wind	WSAvg	8.6	7.9	8.6	9.4	9.9	9.8	9.3	8.7	8.2	7.6	8.0	7.8	7.8
Precipitation	PrecAvg	28.1	1.3	1.5	2.2	2.2	3.5	3.0	2.8	2.2	3.2	2.7	1.8	1.3
	PrecMax	51.7	3.9	6.0	7.7	8.5	7.9	9.1	12.4	11.8	9.2	9.4	5.9	8.6
	PrecMin	11.4	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.1	0.6	0.0	0.0	0.0
	PrecStd	10.7	1.2	1.4	1.6	1.9	2.1	2.5	3.5	2.4	2.1	2.6	1.7	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.6	86.6	88.9	95.8	98.0	101.7	102.8	102.9	100.6	93.4	86.2	81.0
		MCWB	60.5	61.7	65.8	68.7	72.9	74.0	72.4	73.2	71.7	71.7	68.9	64.1
	2%	DB	76.0	80.9	84.5	90.1	94.7	98.6	100.2	100.4	97.0	90.1	82.0	76.2
		MCWB	60.1	62.3	65.0	68.4	72.9	73.9	73.0	73.7	72.4	71.4	66.5	61.7
	5%	DB	71.9	76.6	81.0	86.3	91.7	96.0	98.3	99.0	94.2	87.4	78.8	72.5
		MCWB	58.6	61.1	64.8	68.0	72.9	73.7	73.4	73.9	72.2	70.5	65.1	61.4
	10%	DB	68.1	72.3	77.1	82.9	88.7	93.4	95.9	97.1	91.3	84.5	75.3	69.0
		MCWB	58.0	60.6	63.6	67.2	72.6	74.1	73.8	73.8	72.7	69.1	64.3	59.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.8	69.6	71.3	75.1	78.4	79.5	78.5	78.5	78.3	77.3	73.3	69.7
		MCDB	72.3	75.9	80.2	84.6	88.8	90.4	89.9	92.1	87.5	84.7	79.0	75.0
	2%	WB	65.7	67.8	69.7	73.2	76.9	78.3	77.7	77.6	77.2	75.6	71.1	67.6
		MCDB	70.3	73.7	77.2	82.3	87.1	88.9	89.1	90.3	86.3	82.2	76.9	72.2
	5%	WB	63.5	66.1	68.5	72.0	75.7	77.4	77.0	77.0	76.2	74.5	69.3	65.2
		MCDB	67.6	71.8	75.3	80.4	85.4	87.9	88.0	89.1	85.1	81.2	74.8	69.3
	10%	WB	60.2	64.0	67.1	70.8	74.7	76.5	76.2	76.2	75.4	73.2	67.3	62.0
		MCDB	66.1	69.1	73.4	78.5	83.8	86.5	86.6	87.8	84.0	80.1	73.4	66.8
Mean Daily Temperature Range	5% DB	MDBR	24.7	24.6	23.5	23.5	21.1	21.2	22.0	23.2	22.1	23.0	23.4	24.5
		MCDBR	31.2	29.6	27.6	27.4	25.0	24.8	26.0	26.1	25.7	23.7	25.6	29.3
		MCWBR	18.3	16.4	13.7	11.8	8.3	5.9	5.8	6.2	7.5	9.7	13.9	17.7
	5% WB	MCDBR	20.8	20.9	20.5	19.6	19.5	19.5	20.7	21.8	19.0	17.5	19.4	21.0
		MCWBR	14.7	13.6	11.6	9.6	7.7	6.3	5.8	5.7	6.4	8.2	12.5	15.4
Clear-Sky Solar Irradiance	taub	0.326	0.334	0.355	0.391	0.427	0.418	0.423	0.412	0.404	0.371	0.351	0.321	
	taud	2.498	2.475	2.435	2.349	2.299	2.378	2.394	2.411	2.382	2.463	2.454	2.518	
	Ebn at Noon	290	297	296	287	275	276	274	277	276	280	278	286	
	Edh at Noon	29	32	36	40	42	39	38	37	37	32	30	27	
All-Sky Solar Radiation	RadAvg	976	1160	1449	1715	1823	2105	2118	2029	1655	1365	1038	870	
	RadStd	112	198	151	174	205	153	183	116	162	189	126	117	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.08	+1.26	N/A	N/A	N/A	N/A	N/A	N/A	+441	+347
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157

Nomenclature: See separate page